

Name:Supriya.B Register No:250701760 Experiment Name:Feature Scaling

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import numpy as np
import pandas as pd
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler, MinMaxScaler

df = pd.read_csv('pre-process_datasample.csv')

df['Country'] = df['Country'].fillna(df['Country'].mode()[0])

features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values

age = SimpleImputer(strategy="mean", missing_values=np.nan) salary =
SimpleImputer(strategy="mean", missing_values=np.nan)

features[:, 1:2] = age.fit_transform(features[:, 1:2]) features[:, 2:3] =
salary.fit_transform(features[:, 2:3])

oh = OneHotEncoder(sparse_output=False)
Country = oh.fit_transform(features[:, 0:1])

final_set = np.concatenate((Country, features[:, 1:3]), axis=1)

sc = StandardScaler()
feat_standard_scaler = sc.fit_transform(final_set)
print(feat_standard_scaler)

mms = MinMaxScaler(feature_range=(0, 1))
feat_minmax_scaler = mms.fit_transform(final_set)
print(feat_minmax_scaler)

[[ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 7.58874362e-01
 7.49473254e-01]
 [-8.16496581e-01 -6.54653671e-01 1.52752523e+00 -1.71150388e+00
 -1.43817841e+00]
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 6.63219199e-16]
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 -5.26656882e-01]
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 -1.07356980e+00]
 [ 1.22474487e+00 -6.54653671e-01 -6.54653671e-01 1.34013983e+00
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1.38753832e+00]

[-8.16496581e-01 1.52752523e+00 -6.54653671e-01 1.63077256e+00

1.75214693e+00]

[1.22474487e+00 -6.54653671e-01 -6.54653671e-01 -2.58340208e-01

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[[1. 0. 0. 0.73913043 0.68571429] [0. 0. 1. 0. 0.] [0. 1. 0. 0.13043478

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0.45079365] [1. 0. 0. 0.34782609 0.28571429] [0. 0. 1. 0.51207729

0.11428571] [1. 0. 0. 0.91304348 0.88571429] [0. 1. 0. 1. 1.] [1. 0. 0.

0.43478261 0.54285714]]